

Worksheet: Single-Effect Evaporator

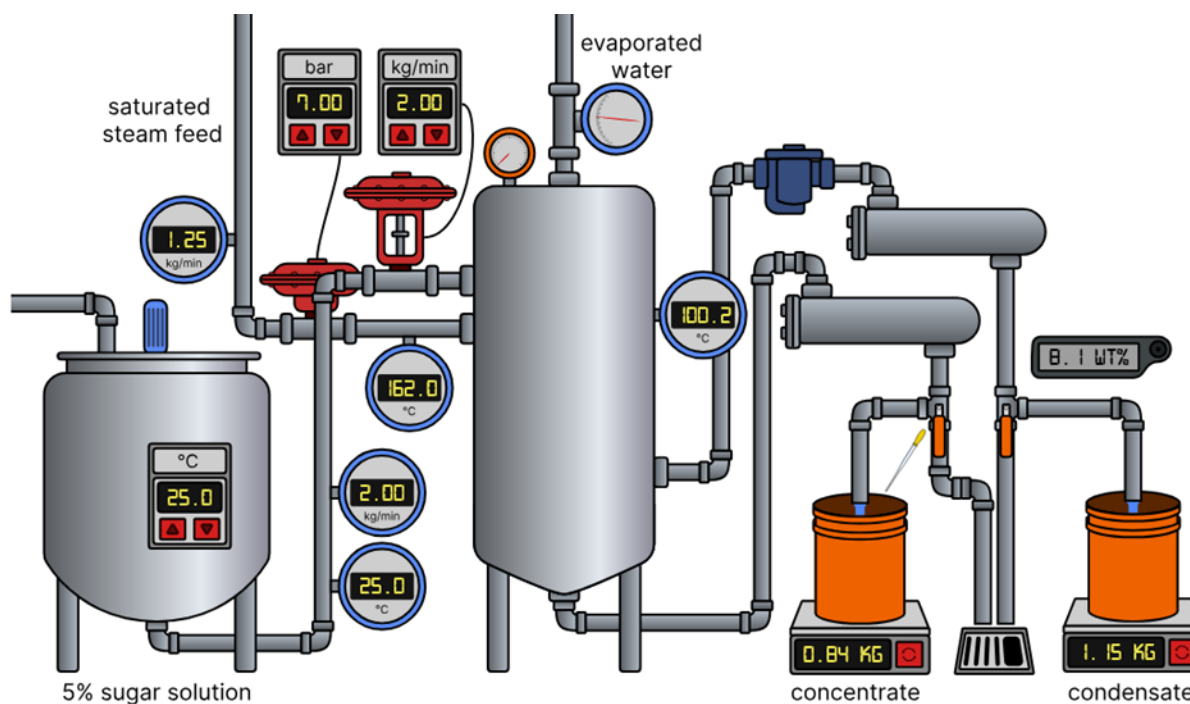
Name(s) _____

This experiment concentrates a dilute sugar solution feed by boiling off the water in a single-effect evaporator using saturated steam as the heat source. Mass and energy balances are carried out. It aims to demonstrate the effects of operating parameters on evaporator efficiency. The evaporator operates at 1.0-bar pressure; the stream pressure can be adjusted, and the pressure determines the steam temperature and flow rate.

Student Learning Objectives

1. Perform mass and energy balances on an evaporative system.
2. Explain the principles of evaporation and concentration in a single-effect evaporator.
3. Explain how pressure and solute concentration affect the boiling temperature of a liquid.
4. Calculate the energy efficiency of a single-effect evaporator.

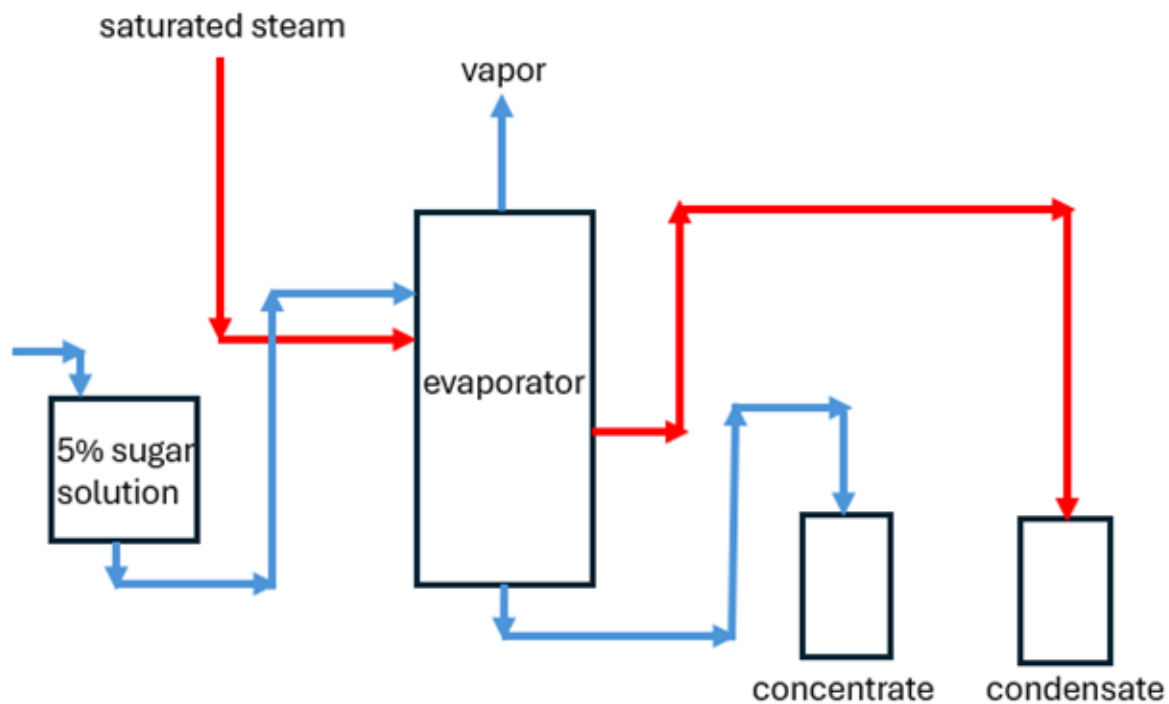
Equipment



- A lab-scale, single-effect evaporator with a 5.0 wt.% sugar solution feed, concentrated sugar solution outlet (concentrate), and an evaporated water outlet.
- The single-effect evaporator has a steam feed, which heats the evaporator contents, and a condensed steam outlet (condensate).
- A pressure controller sets the feed pressure of the saturated steam; this determines the steam temperature and feed rate. A flow meter measures the steam feed rate.
- Flow controllers set the feed rate of the 5.0% sugar solution, and a flow meter measures the feed rate.

- A flow meter measures the rate of evaporated water leaving the evaporator.
- Two containers collect the concentrate and the condensate. The containers are on scales that weight the container contents.
- Thermocouples measure the solution feed temperature, the steam feed temperature, and the evaporator temperature.
- A meter measures the concentration of the concentrate.
- A pressure gauge on the evaporator measures the evaporator pressure which is 1.0 bar.
- A shut off system when the sugar concentration gets too high in the concentrate.

Process flow diagram



Questions to answer before starting the experiment

1. What is the advantage of running the evaporator at a low pressure? How does this affect evaporator temperature?
2. What is the advantage of feeding the sugar solution at higher temperatures?

3. If the steam pressure increases, what happens to the steam temperature if saturated steam is fed to the evaporator?
4. As the sugar concentration in the evaporator increases, what happens to the evaporator temperature?

Notation

- P_{steam} = feed pressure of saturated steam (°C)
 T_{steam} = feed temperature of steam (°C)
 T_f = feed tank temperature (°C)
 T_{evap} = evaporator temperature (°C)
 $\dot{m}_f$ = feed mass flow rate of 5% sugar solution (kg/min)
 $\dot{m}_{vapor}$ = evaporated water mass flow rate (kg/min)
 $\dot{m}_{conc}$ = mass flow rate of concentrate (kg/min)
 $\dot{m}_{steam}$ = mass flow rate of saturated steam (kg/min)
 $\dot{m}_{cond}$ = mass flow rate of condensate (kg/min)
 ΔH_{steam} = heat of vaporization of steam (kJ/kg)
 ΔH_{vap} = heat of vaporization of water at the evaporator temperature (kJ/kg)
 $\dot{Q}$ = rate of heat input to the evaporator (kJ/min)
 Δt = elapsed time (min)
 Δm = change in mass collected in bucket
 C_s = sugar mass fraction (%)

Experiments

Mass balances and energy balances

1. Select the feed pressure of the saturated steam P_{steam} by clicking on the black triangle under the pressure readout. This fixes the steam feed temperature T_{steam} and the steam mass flow rate $\dot{m}_{steam}$. Record these values in Table 1
2. Select the feed tank temperature T_f by clicking on the black triangles below the tank temperature. Record the value in Table 1. The minimum feed temperature is 25°C.
3. Select a feed flow rate of the 5.0 wt.% sugar solution $\dot{m}_f$ by clicking on the black triangle below the flow rate. Record the value in Table 1.
4. Allow the evaporator to reach steady state (this digital experiment accelerates time).
5. Click the orange valve handle that leads to the concentrate bucket. Start a timer and measure the change in mass in the concentrate tank Δm for a time interval Δt . Record the

values in Table 1. Calculate the mass flow rate of the concentrate $\dot{m}_{conc}$ and record the value in Table 1.

6. Mouse over the meter at the top of the evaporator to see the evaporated water flow rate $\dot{m}_{vapor}$. Record the value in Table 1.
7. Measure the concentration (wt.%) of the concentrate by clicking the meter that is above the condensate bucket. Record the sugar concentration as a mass fraction C_s in Table 1.
8. Do a total mass balance: in - out = $\dot{m}_f - \dot{m}_{conc} - \dot{m}_{vapor}$
9. Since the system is at steady state, this should be zero. Calculate the percent difference and record in Table 1:

$$\% \text{ difference} = \left(\frac{\dot{m}_f - \dot{m}_{conc} - \dot{m}_{vapor}}{\dot{m}_f} \right) \times 100$$

10. Calculate the sugar feed rate $0.05\dot{m}_f$ and the sugar flow rate out $C_s\dot{m}_{conc}$ and record those values in Table 1. Do a sugar mass balance: in - out = $0.05\dot{m}_f - C_s\dot{m}_{conc}$ and calculate the percent difference and record in Table 1.

$$\% \text{ difference} = \left(\frac{0.05\dot{m}_f - C_s\dot{m}_{conc}}{0.05\dot{m}_f} \right) \times 100$$

Table 1 Mass balances P_{steam} _____ T_{steam} _____ T_f _____

Concentrate		Total mass flow rates (kg/min)			Total		Sugar mass flow rates (kg/min)		Sugar
Δt (s)	Δm (kg)	$\dot{m}_f$	$\dot{m}_{conc}$	$\dot{m}_{vapor}$	% difference	C_s	$0.05\dot{m}_f$	$C_s\dot{m}_{conc}$	% difference

11. Discuss the mass balances and reasons that the % difference deviates from zero.

12. Click the orange valve handle that leads to the condensate bucket. Start a timer and measure the change in mass Δm in the condensate bucket for a time interval Δt . Record the values in Table 2. Calculate the mass flow rate of the condensate $\dot{m}_{cond}$ and record it in Table 2. Add the steam feed rate $\dot{m}_{steam}$ to Table 2 and calculate the percent difference $[(\dot{m}_{steam} - \dot{m}_{cond})/\dot{m}_{steam}] \times 100$.

Table 2 Steam condensate

Steam feed			Steam condensate			
P_{steam} (bar)	T_{steam} (°C)	$\dot{m}_{steam}$ (kg/min)	Δt (s)	Δm (kg)	$\dot{m}_{cond}$ (kg/min)	% Difference

Steam economy

Use the same feed temperature and feed flow rate as in Experiment 1 and record these values in Table 3. Record $\dot{m}_{vapor}$ and $\dot{m}_{steam}$ in Table 3 at four steam pressures between 2 and 20 bar at steady-state conditions. Calculate the steam economy for each pressure and record in Table 3.

$$steam\ economy = \frac{\dot{m}_{vapor}}{\dot{m}_{steam}}$$

Table 3 Mass balances T_f _____ $\dot{m}_f$ _____

P_{steam} (bar)	T_{steam} (°C)	$\dot{m}_{vapor}$	$\dot{m}_{steam}$	$\frac{\dot{m}_{vapor}}{\dot{m}_{steam}}$

Plot steam economy versus steam temperature. What do you observe?

Questions to answer

1. How is the evaporator temperature related to the concentrate concentration?
2. Why does increasing steam pressure usually increase evaporation rate?
3. Why is evaporator pressure more important than steam pressure in determining the boiling temperature of the liquid?
4. Where might these measurements have errors?
5. How does increasing the feed temperature affect the outlet compositions and flow rates?
6. What safety precautions would you take to conduct this experiment in the laboratory?
7. How would you use three evaporators in series? What would be the advantage of doing this?